Lee Sungyo

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EDUCATION

University of Illinois Urbana-Champaign **Expected May 2028**
B.S. in Electrical Engineering *GPA: 3.82/4.00*

- **Current Coursework:** ECE 210 (Analog Signals), ECE 313 (Probability), ECE 385 (Digital Systems Lab)
- **Completed Coursework:** Data Structures, Discrete Mathematics, Linear Algebra, C/C++ Programming
- **Technical Interests:** GPU programming, computer graphics, embedded systems, and performance analysis

SELECTED ENGINEERING PROJECTS

GPU CNN Inference and Kernel Fusion *CUDA C++, NVIDIA A40*

- Implemented batched CUDA convolution for modified LeNet-5 on Fashion-MNIST using im2col input unrolling and shared-memory tiled matrix multiplication.
- Fused unrolling, matrix multiplication, and output permutation into one kernel; profiled CPU/GPU variants with gprof, Nsight Systems, and Nsight Compute.
- Matched CPU-reference accuracy of 87.14% on 10,000 images; measured 56.70 ms combined kernel time for the direct-convolution baseline on NVIDIA A40.

USB-Controlled HDMI Graphics System *C, SystemVerilog, MicroBlaze, SPI*

- Built a keyboard-controlled graphics demo on a MicroBlaze FPGA SoC, processing USB input to update objects on an HDMI display.
- Implemented C routines for single-register and multi-byte transfers to a MAX3421E USB host controller over SPI.
- Used UART keycode output and hardware display tests to debug keyboard input and graphics behavior.

AXI4-Lite Color Text Controller *SystemVerilog, BRAM, Vivado, Vitis*

- Developed an 80×30 HDMI text controller with dual-port BRAM VRAM, font-ROM rendering, and per-character foreground/background colors.
- Adapted AXI4-Lite handshaking for synchronous BRAM latency; exposed VRAM, a programmable 16-color palette, and synchronization registers through memory-mapped I/O.
- Implemented C palette routines; verified AXI readback in simulation and frame-synchronized color animation on FPGA.

RISC-V Kernel and Filesystem *C, RISC-V, QEMU*

- Extended an RV64 educational kernel with a write-back LRU block cache and NGFS file operations, including file creation, resizing, deletion, and per-file locking.
- Implemented ELF validation/loading with segment initialization and entry-point checks; ran supported game executables in QEMU and passed filesystem/cache tests.

WORK EXPERIENCE

ISU Petasys Corp. *PCB Production Operator* **May 2026 – July 2026**

- Operated PCB inner-layer production equipment on 12-hour rotating shifts; inspected panels for opens, shorts, contamination, and surface defects.

Republic of Korea Army *Cybersecurity Analyst* **October 2024 – April 2026**

- Investigated Splunk alerts, network traffic, and system logs across corps-level systems; supported incident response and remediation.
- Administered Linux servers, user access, Network Access Control (NAC), and firewall policies for military information systems.
- Wrote Bash and Python scripts to automate system administration, backups, and file-integrity verification.

TECHNICAL SKILLS

Programming: C, C++, CUDA C++, Python, SystemVerilog, RISC-V Assembly
Tools: Linux, Git, Nsight Systems/Compute, gprof, Vivado, Vitis, Bash, MATLAB, Splunk, LaTeX
Hardware: FPGA verification, RTL/FSM design, AXI4-Lite, SPI, UART, memory-mapped I/O
Languages: Korean, Chinese